TEST PLANS AND RESULTS:

We initially had a lot of ideas or components to be integrated into this project of ours, Spin Shot. But due to complexities, we had to skip them to prioritise the basic functioning.

Such ideas consisted of having a buzzer and an IR sensor integrated into the target box, where the sensor would detect the arrow entering through the cut holes and trigger the buzzer. But during testing, the IR sensor had a vast angular range, and due to light and obstacles bouncing off everywhere, it kept getting inputs, ruining the required readings.

We had a lot of issues with wire connectivity. The first one was the servo motors, which kept faltering. And after analysing every component connected to it, we came to know that there were issues in the jumper wires connecting them. So, we changed entire sets of wires and got them to work properly.

The second issue, which is still in question, is the working of the DC motor, which has been very difficult to debug. The motor runs, then stops, then runs again. And sometimes, we found it working even with the wire connectivity with the female jack detached, confusing us even more. So, we have continued working with an unpredictably functioning DC motor.

Our first plan to 3D print the gun failed due to the weakness of the model. The scale and density infill were very weak, so when I added the rubber band mechanism, the model broke due to the force of the rubber. So, we fixed this by mimicking the way the model worked, using MDF, which proved to be much stronger and more durable.